AGE 310: Instrumentation and Measurements

Dr. Kadeghe Fue, Department of Agricultural Engineering, SUA

Morogoro, Tanzania

Abstract

Lecture 13 provides an in-depth analysis of Unmanned Aerial Vehicles (UAVs) specifically for agriculture. We classify drones, analyze their core electronic components (Flight Controllers, IMU, RTK-GNSS), and explore specialized sensing payloads like Multispectral and Thermal cameras. Crucially, the lecture covers the practical and legal requirements for operating drones in Tanzania under the TCAA regulations.

Lecture 13: Agricultural Drones (UAVs)

13.1 Drone Classification and Structure

13.1.1 Types of Agricultural Drones

1. **Multi-Rotor (Rotary Wing):** Uses multiple vertical propellers (quadcopters, hexacopters, octocopters).
    - **Pros:** Vertical Take-Off and Landing (VTOL), high maneuverability (can hover), excellent for precision spot spraying and close inspection.
    - **Cons:** Low endurance (short battery life, typically < 30 minutes), limited range.
2. **Fixed-Wing:** Resembles a traditional aircraft. Uses wings for lift and a propeller for forward thrust.
    - **Pros:** High endurance (1 hour $+$), capable of mapping large areas (e.g., 1000 ha) in a single flight.
    - **Cons:** Cannot hover, requires a launch system (catapult or hand throw) and landing space.

13.1.2 Core Electronic Components

The functionality of a drone relies on integrating multiple instrumentation systems:

- **Flight Controller (FC):** The brain. Contains the central MCU (Microcontroller) which runs the operating system (OS) and the PID algorithms (Lecture 11) to stabilize the drone.
- **IMU (Inertial Measurement Unit):** Consists of accelerometers and gyroscopes (Piezoelectric, Lecture 5) that measure the drone's orientation (pitch, roll, yaw) 100s of times per second. Essential for PID stability.
- **ESC (Electronic Speed Controller):** Interfaces the low-power FC signal to the high-power Brushless DC motors (BLDC, Lecture 7), precisely controlling the RPM of each propeller for controlled movement.
- **GNSS/RTK Module:** Provides positioning data. RTK (Real-Time Kinematic) systems use ground reference stations to achieve centimeter-level accuracy, crucial for auto-spraying over crops.

13.2 Sensing Payloads

Drones are primarily aerial sensor platforms. The choice of sensor dictates the application.

13.2.1 RGB (Visible Light) Cameras

- **Function:** Standard visible light ($400 - 700$ nm) image capture.
- **Applications:** Visual scouting, counting plant population, monitoring emergence, and generating high-resolution orthomosaic maps.

13.2.2 Multispectral Imaging

These cameras capture data outside the visible spectrum, particularly in the Near-Infrared (NIR) region.

- **Principle:** Healthy plants strongly absorb red light and strongly reflect NIR light due to high chlorophyll content. Stressed or dead plants reflect less NIR.
- **NDVI (Normalized Difference Vegetation Index):** The key metric derived from multispectral data.

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

- **Interpretation:** NDVI values range from -1 to $+1$. Healthy vegetation is typically > 0.6. This is used to map crop health, disease hotspots, and nutrient deficiencies.

13.2.3 Thermal and LiDAR Sensing

- **Thermal Cameras (IR):** Measure temperature based on emitted infrared radiation.
  - **Applications:** Detecting plant water stress (stressed plants are hotter due to closed stomata), checking heat leakage in grain storage silos, and livestock monitoring.
- **LiDAR (Light Detection and Ranging):** Uses laser pulses to measure distance and generate precise 3D point clouds.
  - **Applications:** Generating detailed Digital Elevation Models (DEM) for farm drainage planning, and calculating tree canopy volume for optimized fruit spraying.

13.3 Regulatory Framework (Tanzania Context)

Operating UAVs in Tanzania is strictly governed by the Tanzania Civil Aviation Authority (TCAA). Every Agricultural Engineer must be aware of these regulations.

13.3.1 Key TCAA Requirements

1. **Registration and Permits:** All drones, regardless of size, must be registered with the TCAA. An import permit must be obtained before the drone physically enters the country.
2. **Weight Restrictions:** Drones weighing over 7 kg (which includes most professional spraying drones) require special clearance from the Ministry of Defense and National Service.
3. **Flight Rules:**
    - **VLOS (Visual Line of Sight):** The operator must maintain continuous unaided visual contact with the UAV at all times.
    - **Maximum Altitude:** 400 ft (120 m) above ground level.
    - **No-Fly Zones:** Strict prohibition near airports (e.g., Julius Nyerere, Kilimanjaro) and restricted military areas.

Consequence: Non-compliance can lead to confiscation, heavy fines, and imprisonment.

13.3.2 Flight Planning

To conduct efficient and legal missions, specialized software (e.g., DJI Pilot, Pix4D) is used for automated flight path generation.

1. **Boundary Mapping:** Defining the exact perimeter of the field using GPS coordinates.
2. **Overlap Setting:** Specifying the percentage overlap (e.g., 70% front overlap, 60% side overlap) to ensure continuity for creating orthomosaic maps (*stitching*).
3. **GSD (Ground Sample Distance):** The distance between the centers of two consecutive pixels measured on the ground. Determined by camera resolution and flight altitude. A lower GSD (e.g., 2 cm/pixel) means higher resolution and lower altitude.

13.4 Agricultural Applications (Tanzania Context)

13.4.1 Sugarcane Ripener Application

- **Problem:** Sugarcane ripeners must be applied precisely in the weeks leading up to harvest.
- **Solution:** Large multi-rotor drones with high payload capacity apply the ripener over dense, tall cane fields, which are difficult for traditional ground sprayers to penetrate.

13.4.2 Pest and Disease Hotspot Detection

- **Process:** NDVI maps from drone scouting identify areas of crop stress (low NDVI).
- **Action:** Field workers or spot spraying drones are then deployed only to those precise *hotspots*, minimizing the blanket use of pesticides/fungicides.

Calculation Example 1: Spraying Coverage and Time

A large spraying drone carries 10 L of pesticide. The required application rate is 20 L/ha. The drone flies at 5 m/s with a 6 m spray width.

1. Calculate the area covered per 10 L tank load in hectares (ha).
2. Calculate the total time taken to empty the tank (flight time).

Solution:

1. **Area Covered (A):**

$$A = \frac{\text{Tank Capacity}}{\text{Application Rate}} = \frac{10 \text{ L}}{20 \text{ L/ha}} = 0.5 \text{ ha}$$

2. **Flow Rate (Q):** Must be calculated to empty 10 L while covering 0.5 ha. Area Covered per second: 5 m/s $\times$ 6 m $= 30$ m^2/s. Area in ha/s: 30 m^2/s $\times \frac{1 \text{ ha}}{10{,}000 \text{ m}^2} = 0.003$ ha/s.

   Time to cover 0.5 ha: $\frac{0.5 \text{ ha}}{0.003 \text{ ha/s}} \approx 166.7$ seconds.

   Verification using flow rate: Required flow rate (Q): 0.003 ha/s $\times 20$ L/ha $= 0.06$ L/s. Total Time: $\frac{\text{Volume}}{\text{Flow Rate}} = \frac{10 \text{ L}}{0.06 \text{ L/s}} \approx 166.7$ seconds ≈ 2 minutes, 47 seconds.

13.5 Summary of Key Terms (Swahili/Context)

- **UAV:** Ndege Isiyo na Rubani (Unmanned Aerial Vehicle).
- **VTOL:** Kuruka na Kutua kwa Wima (Vertical Take-Off and Landing).
- **GSD:** Umbali wa Sampuli ya Ardhi (Ground Sample Distance - map resolution).
- **NDVI:** Kielezo cha Mimea (Vegetation Health Index).
- **RTK-GNSS:** RTK-GNSS (Centimeter-accurate GPS).
- **TCAA:** Mamlaka ya Usafiri wa Anga Tanzania (Civil Aviation Authority).
- **VLOS:** Kuona kwa macho (Visual Line of Sight).
- **Orthomosaic:** Ramani Imetengenezwa kwa Picha (Stitched, corrected aerial map).

Lecture 13: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. Differentiate between a *Multi-Rotor* and a *Fixed-Wing* drone based on flight endurance and maneuverability. Which type is preferred for large-scale mapping (e.g., 1000 ha of wheat)?
2. State the function of the *IMU* (Inertial Measurement Unit) and the *ESC* (Electronic Speed Controller) within the drone's structure.
3. Define NDVI (*Normalized Difference Vegetation Index*). What specific wavelength band (light color) do healthy plants reflect strongly to yield a high NDVI value?
4. According to TCAA regulations, state the maximum allowed flying altitude for a drone in Tanzania and the meaning of VLOS (*Visual Line of Sight*).
5. What is the purpose of *LiDAR* sensing in agricultural applications? Provide one specific application where LiDAR is superior to standard RGB camera data.

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **NDVI and Crop Stress:** An RGB-NIR sensor measures 200 counts in the Red band and 550 counts in the NIR band for a maize field.
   - Calculate the NDVI value for this sample.
   - If the ideal NDVI is 0.7, what does the calculated value suggest about the crop's health (e.g., nutrient deficiency or water stress)?
7. **Ground Sample Distance (GSD):** A drone camera has a sensor width of 36 mm and a focal length of 18 mm. The orthomosaic map requires a GSD of 5 cm/pixel. Calculate the required flight altitude (H) in meters to achieve this GSD, assuming a sensor pixel pitch of 5 µm. (Formula: $GSD = \frac{H \cdot \text{pixel size}}{\text{focal length}}$).
8. **TCAA Weight Clearance:** A professional spraying drone has a net weight of 12 kg and carries a 10 L tank of pesticide (density ≈ 1 kg/L). Calculate the total take-off weight (TOW). Explain the legal consequence in Tanzania of flying this drone without additional clearance.
9. **Flight Planning Overlap:** Explain the necessity of setting a high percentage overlap ($> 60\%$) during drone mission planning for mapping. What specific post-processing error does insufficient overlap cause?
10. **RTK-GNSS Accuracy for Sprayers:** Explain why 2 cm accuracy (RTK-GNSS) is mandatory for autonomous spraying drones but unnecessary for simple scouting drones. Relate your answer to the cost of pesticides and chemical drift.
11. **Flow Rate and Coverage Time:** A small drone carries 5 L and sprays at 15 L/ha. The battery life allows 15 minutes of flight time.
    - Calculate the area (in ha) the drone can cover per battery charge.
    - What is the drone's effective work rate in ha/min?

12. **Fixed-Wing Cost Justification:** If a fixed-wing drone costs 4 times more than a multi-rotor drone, justify the investment for a large-scale farm ($10,000$ ha of maize) based on coverage and endurance benefits.

13. **Data Storage Overload:** A drone mapping mission generates 500 high-resolution images, averaging 10 MB each. Calculate the total data size generated in Gigabytes (GB). Explain why storing and processing this volume of data is a major challenge for SUA researchers.

14. **Thermal Imaging Application:** Explain how a thermal camera payload can be used to detect water stress in a field of beans, focusing on the physiological mechanism that causes the temperature difference.

15. **Propeller RPM Control:** In a quadcopter, explain why the FC must continuously adjust the RPM of all four motors via ESCs (using PID control) to maintain a simple hover, rather than just keeping them at a constant speed.